

Question Number	Answer	Additional Guidance	Mark
3(a)	There is no (significant) difference between the surface area of toepads before and after the hurricane	Candidates can express this in different ways A and B must be qualified	(1)

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3(b)	- suitable table format with correct column headings and units (1) - all data correctly entered (1) - means correctly calculated (1)	Example table <table><tr><th colspan="2">area of toepad /mm²</th></tr><tr><th>sample A</th><th>Sample B</th></tr><tr><td>0.9</td><td>0.8</td></tr><tr><td>1.3</td><td>1.3</td></tr><tr><td>1.1</td><td>1.5</td></tr><tr><td>1.2</td><td>1.8</td></tr><tr><td>1.3</td><td>1.3</td></tr><tr><td>1.5</td><td>1.5</td></tr><tr><td>1.6</td><td>1.6</td></tr><tr><td>0.8</td><td>1.9</td></tr><tr><td>0.7</td><td>1.8</td></tr><tr><td>Mean 1.16</td><td>Mean 1.50</td></tr></table> Or 1.2 and 1.5 A third column negates MP1	area of toepad /mm ²		sample A	Sample B	0.9	0.8	1.3	1.3	1.1	1.5	1.2	1.8	1.3	1.3	1.5	1.5	1.6	1.6	0.8	1.9	0.7	1.8	Mean 1.16	Mean 1.50	(3)
area of toepad /mm ²																											
sample A	Sample B																										
0.9	0.8																										
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3(c)	- bar graph with linear scale starting at zero and axes labelled (1) - means plotted correctly (1) - range bars plotted correctly (1)	ALLOW ECF from 3bi mean (surface) area (of toepads) /mm² A B	(3)

Question Number	Answer	Additional Guidance	Mark
3(d)(i)	- correct numerator (1) - correct substitution of given $(S_A)^2$ and $(S_B)^2$ divided by 9/denominator (1) - correct answer (1)	ECF for any mean values allow use of 1.16 - 1.50 numerator = 0.1527 If $(S_A)^2$ and $(S_B)^2$ squared again then use denominator 0.0497 only MP1 and 3 $t = 1.96$ Correct answer gains 3 marks Ignore minus sign $t = 2.225/2.23$ if 1.16 mean used	(3)

Question Number	Answer	Additional Guidance	Mark
3(d)(ii)	- the calculated value of t (1.96) is less than the critical value of 2.12 (1) - therefore accept the null hypothesis there is no difference between the surface area (of toepads) before and after the hurricane (1)	<i>Ignore negative t values so do not award MP1</i> Allow ECF from 3di if t value more than 2.12 allow converse statements/ref to 16 D of Fonly Allow sample A and sample B	(2)

Question Number	Answer	Additional Guidance	Mark
3(e)	An answer that includes two of the following points: • repeat after each hurricane (1) • measure hind limb toepads (1) • repeat in other {locations/areas} (1) • other species (of lizard) (1)	ignore repeat the expt	(2)

(Total for question 3 = 14 marks)

